

Homework (Jalapeno)

- Q1.** The ancient Egyptians used $\sqrt{2}$ to represent the diagonal of a square. Rationalize the denominator of $\frac{1}{\sqrt{2}}$.
- (A) $\frac{\sqrt{2}}{2}$
(B) $\frac{2}{\sqrt{2}}$
(C) $\frac{1}{2}$
(D) $\sqrt{2}$
(E) 2
- Q2.** The ancient Babylonians used $\sqrt{3}$ to represent the diagonal of a rectangle. Rationalize the denominator of $\frac{2}{\sqrt{3}}$.
- (A) $\frac{2\sqrt{3}}{3}$
(B) $\frac{\sqrt{3}}{2}$
(C) $\frac{2}{3}$
(D) $\sqrt{3}$
(E) 2
- Q3.** The ancient Greeks used $\sqrt{5}$ to represent the golden ratio. Rationalize the denominator of $\frac{1}{\sqrt{5}}$.
- (A) $\frac{\sqrt{5}}{5}$
(B) $\frac{5}{\sqrt{5}}$
(C) $\frac{1}{5}$
(D) $\sqrt{5}$
(E) 5
- Q4.** The Mayans used $\sqrt{6}$ to represent the length of a bridge. Rationalize the denominator of $\frac{3}{\sqrt{6}}$.
- (A) $\frac{3\sqrt{6}}{6}$
(B) $\frac{\sqrt{6}}{3}$
(C) $\frac{3}{6}$
(D) $\sqrt{6}$
(E) 6
- Q5.** Archimedes used $\sqrt{7}$ to calculate the area of a circle. Rationalize the denominator of $\frac{2}{\sqrt{7}}$.
- (A) $\frac{2\sqrt{7}}{7}$
(B) $\frac{\sqrt{7}}{2}$
(C) $\frac{2}{7}$
(D) $\sqrt{7}$
(E) 7
- Q6.** Euclid used $\sqrt{8}$ to represent the length of a rectangle. Rationalize the denominator of $\frac{1}{\sqrt{8}}$.
- (A) $\frac{\sqrt{8}}{8}$
(B) $\frac{\sqrt{2}}{4}$
(C) $\frac{1}{8}$
(D) $\sqrt{8}$
(E) 8
- Q7.** The ancient Chinese used $\sqrt{9}$ to represent the area of a square. Rationalize the denominator of $\frac{3}{\sqrt{9}}$.
- (A) $\frac{3}{3}$
(B) $\frac{\sqrt{9}}{3}$
(C) $\frac{3}{9}$
(D) $\sqrt{9}$
(E) 9
- Q8.** The ancient Indians used $\sqrt{10}$ to represent the length of a bridge. Rationalize the denominator of $\frac{2}{\sqrt{10}}$.
- (A) $\frac{2\sqrt{10}}{10}$
(B) $\frac{\sqrt{10}}{2}$
(C) $\frac{2}{10}$
(D) $\sqrt{10}$
(E) 10

Open Ended Questions (FRQ)

- Q9.** Rationalize the denominator of $\frac{5}{\sqrt{12}}$ and simplify. Show your work.
- Q10.** Rationalize the denominator of $\frac{4}{\sqrt{15}}$ and simplify. Show your work.

Chef's Tips

- Tip 1: Remind students to simplify after rationalizing the denominator.
- Tip 2: Encourage students to check their work.
- Tip 3: Provide extra support for students who struggle with radicals.